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Class 11 Physics | Chapter 5

Laws of Motion

Complete study notes covering Newton's Laws, friction, circular motion, and key formulas — designed for Class 11 students.

CHAPTER 5 — OVERVIEW

What We'll Study

Chapter 5 — **Laws of Motion** — is one of the most fundamental chapters in Class 11 Physics. It builds the foundation for understanding how forces affect the motion of objects.

01

Newton's Laws of Motion

First, Second, and Third Law with real-world examples

02

Force & Inertia

Types of forces, inertia, and momentum

03

Friction

Static, kinetic, and rolling friction with laws

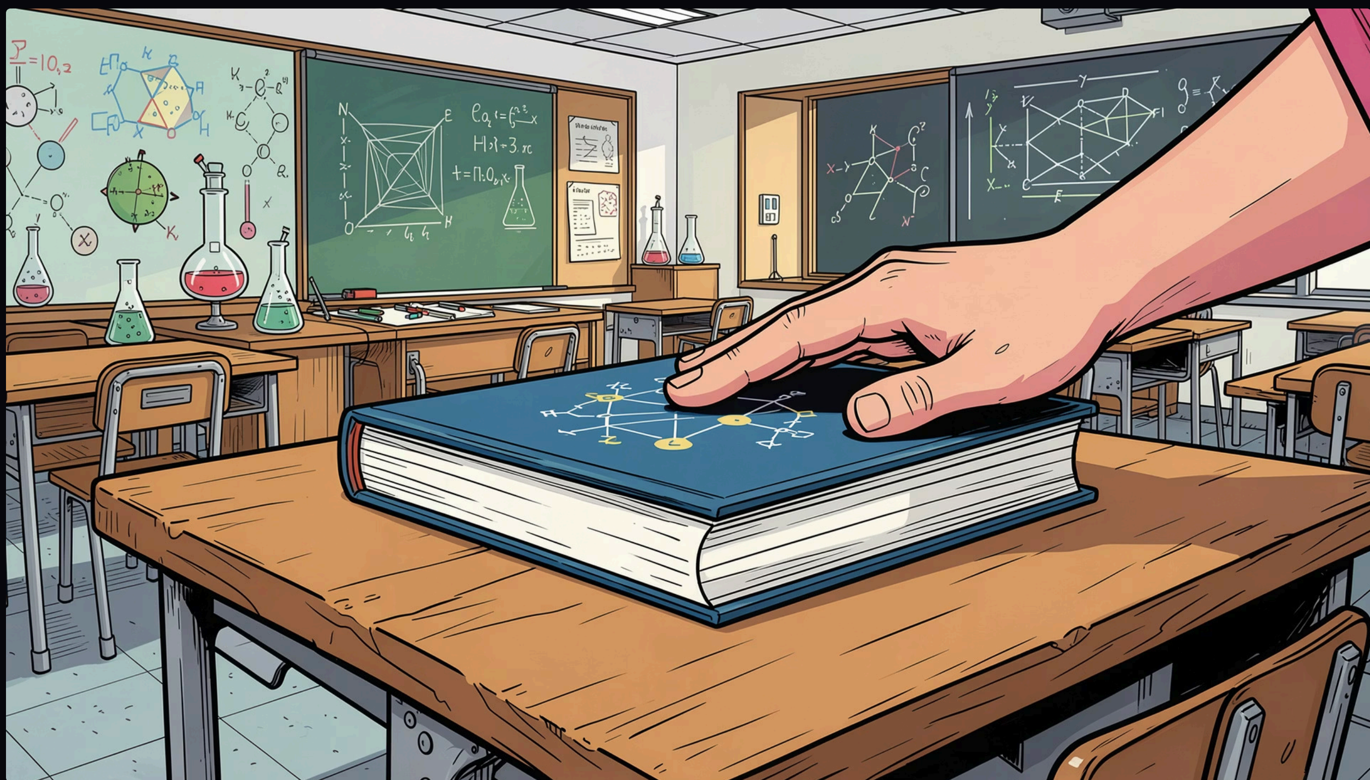
04

Circular Motion

Centripetal force, banking of roads, and applications

NEWTON'S FIRST LAW

Law of Inertia



Statement

"An object remains in its state of rest or of uniform motion in a straight line unless compelled to change that state by an external unbalanced force."

Key Concepts

- **Inertia:** The tendency of an object to resist change in its state of motion
- **Inertia of Rest:** Object at rest stays at rest (e.g., dust on a beaten carpet)
- **Inertia of Motion:** Object in motion stays in motion (e.g., passenger jerks forward when bus stops)
- **Inertia of Direction:** Object resists change in direction (e.g., mud flying tangentially from a rotating wheel)

Inertia is measured by mass. Greater the mass, greater the inertia.

NEWTON'S SECOND LAW

Force, Mass & Acceleration

Statement

"The rate of change of momentum of a body is directly proportional to the applied force and takes place in the direction in which the force acts."

Mathematical Form

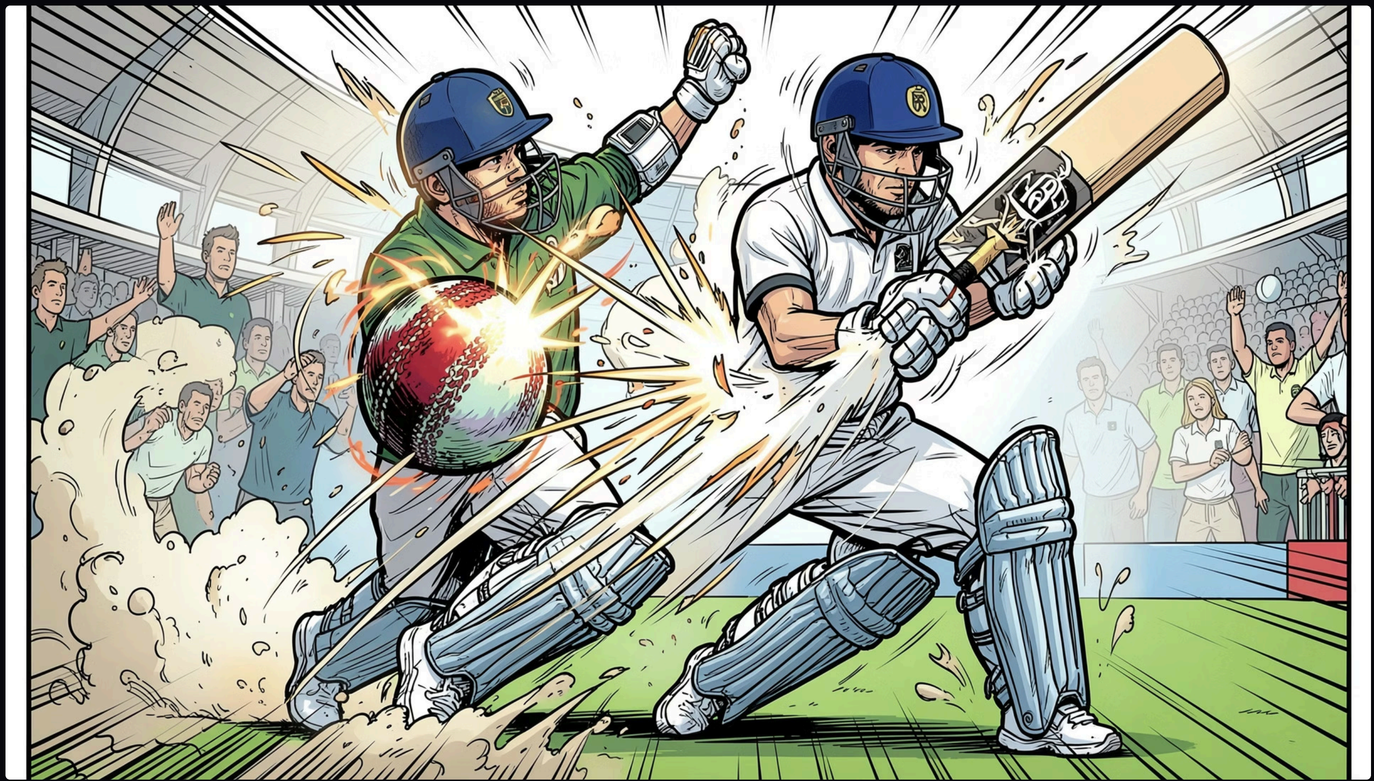
$$F = ma$$

$$p = mv \quad (\text{Momentum})$$

$$F = \frac{dp}{dt}$$

Key Points

- Force is a **vector quantity** (SI unit: Newton, N)
- Acceleration is in the **same direction** as the net force
- $1\ \text{N} = 1\ \text{kg} \cdot \text{m/s}^2$
- If **F = 0**, then **a = 0** (consistent with First Law)



NEWTON'S THIRD LAW

Action & Reaction

Statement

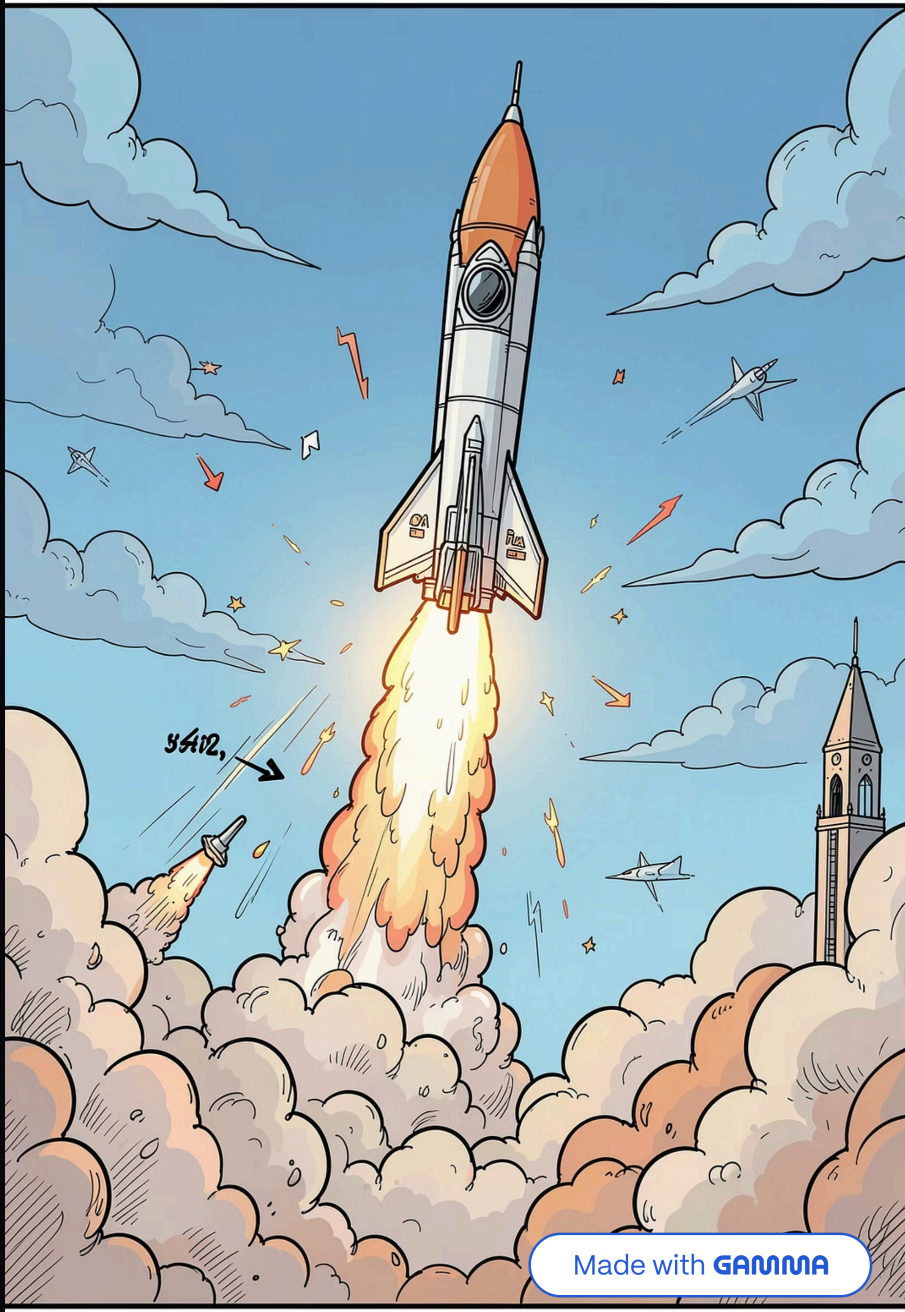
"To every action, there is always an equal and opposite reaction."

Key Rule

Action and reaction forces act on **different bodies** — they never cancel each other.

Examples

- Walking: foot pushes ground backward, ground pushes foot forward
- Rocket: gas expelled downward, rocket moves upward
- Swimming: hand pushes water backward, body moves forward



CONSERVATION LAWS

Conservation of Momentum

Law Statement

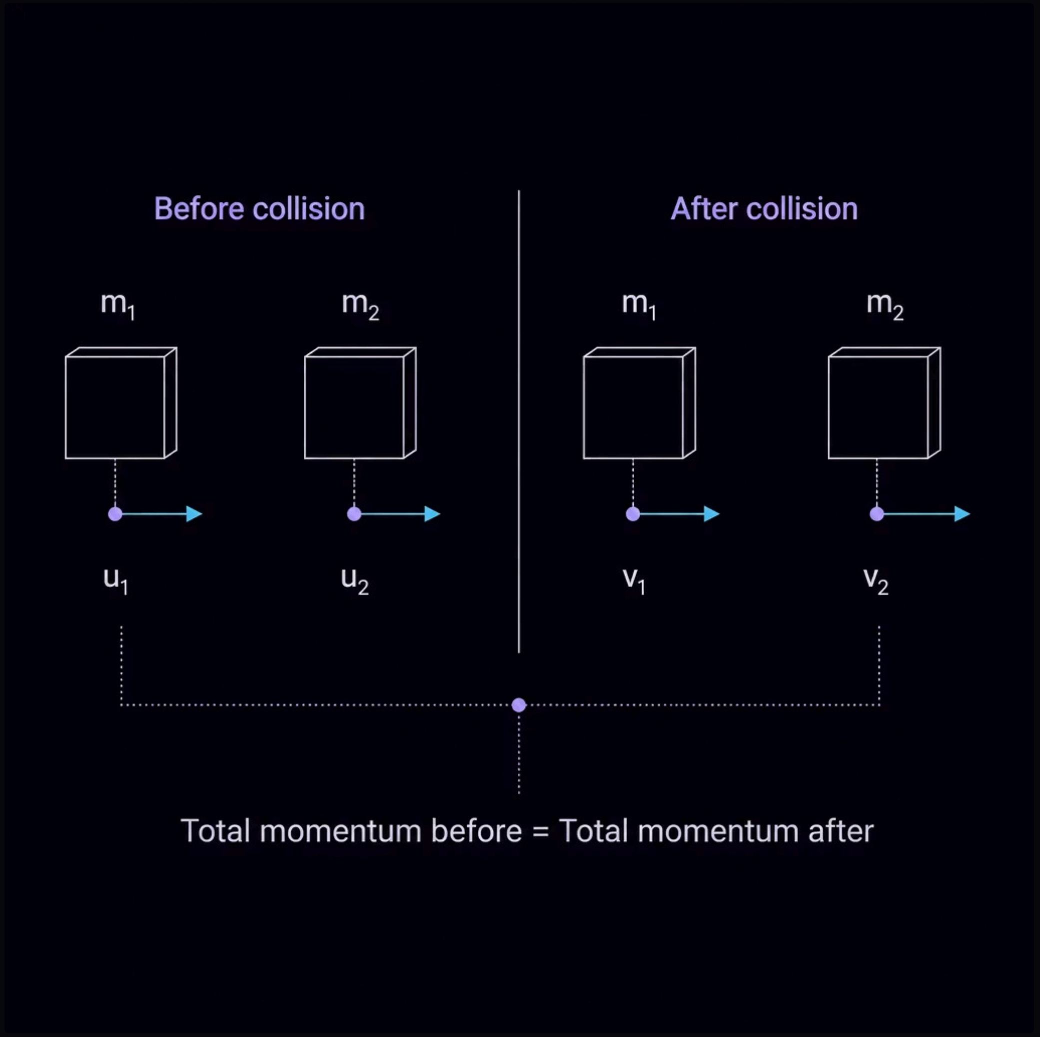
In the absence of external forces, the **total momentum of a system remains constant.**

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

Applications

- **Recoil of a gun:** Gun moves backward when bullet is fired forward
- **Collision of objects:** Two bodies colliding in isolation
- **Rocket propulsion:** Expelling mass to gain forward momentum

Derived directly from Newton's Second and Third Laws.



FRICTION

Types & Laws of Friction

Static Friction

Opposes impending motion. Self-adjusting force up to a maximum value $f_s^{max} = \mu_s N$

Kinetic Friction

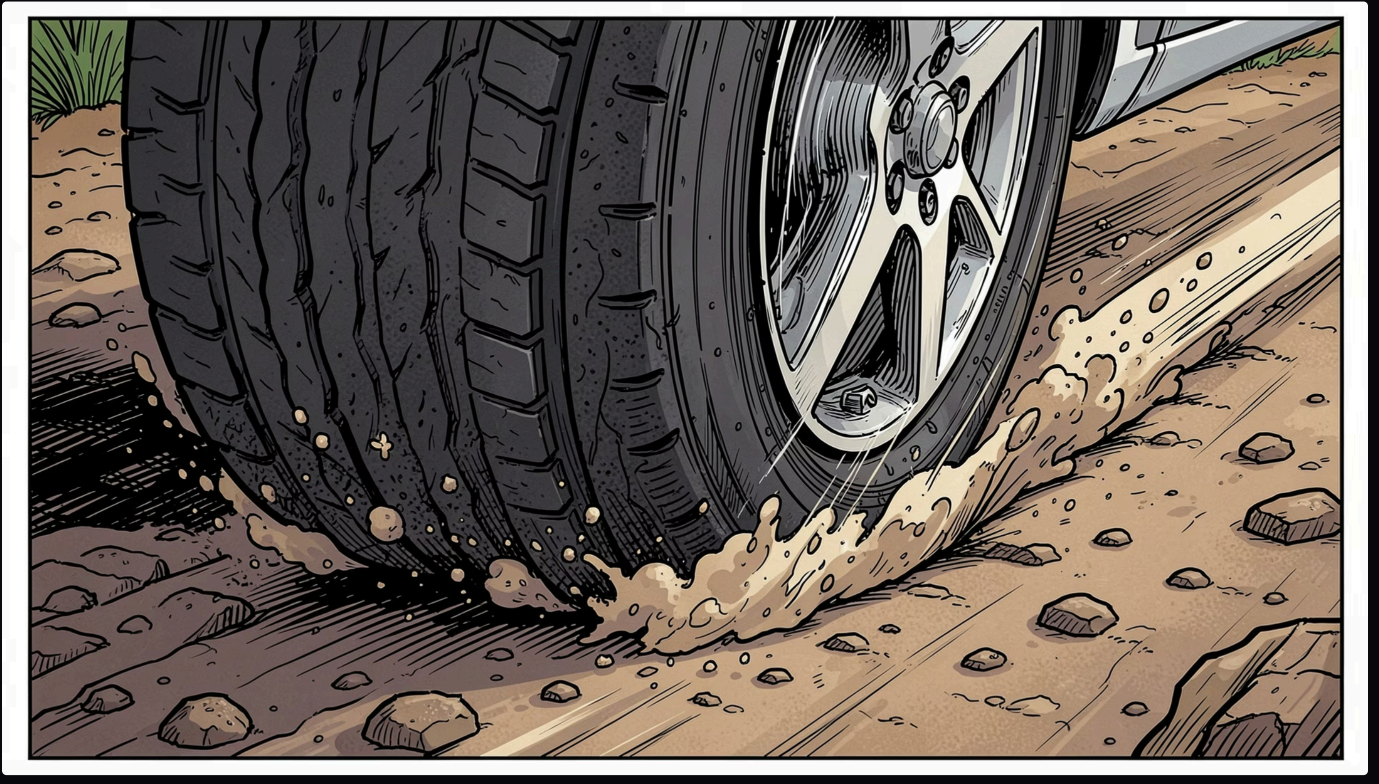
Opposes actual motion. $f_k = \mu_k N$. Always less than maximum static friction.

Rolling Friction

Opposes rolling motion. Much smaller than sliding friction. Used in ball bearings and wheels.

Laws of Friction

- Friction is **independent of area of contact**
- Proportional to **normal reaction (N)**
- $\mu_s > \mu_k$ (static coefficient is greater)
- Angle of friction: $\tan\theta = \mu$



CIRCULAR MOTION

Centripetal Force & Banking

Centripetal Force

Force directed toward the center of the circular path.

$$F_c = \frac{mv^2}{r} = mr\omega^2$$

Banking of Roads

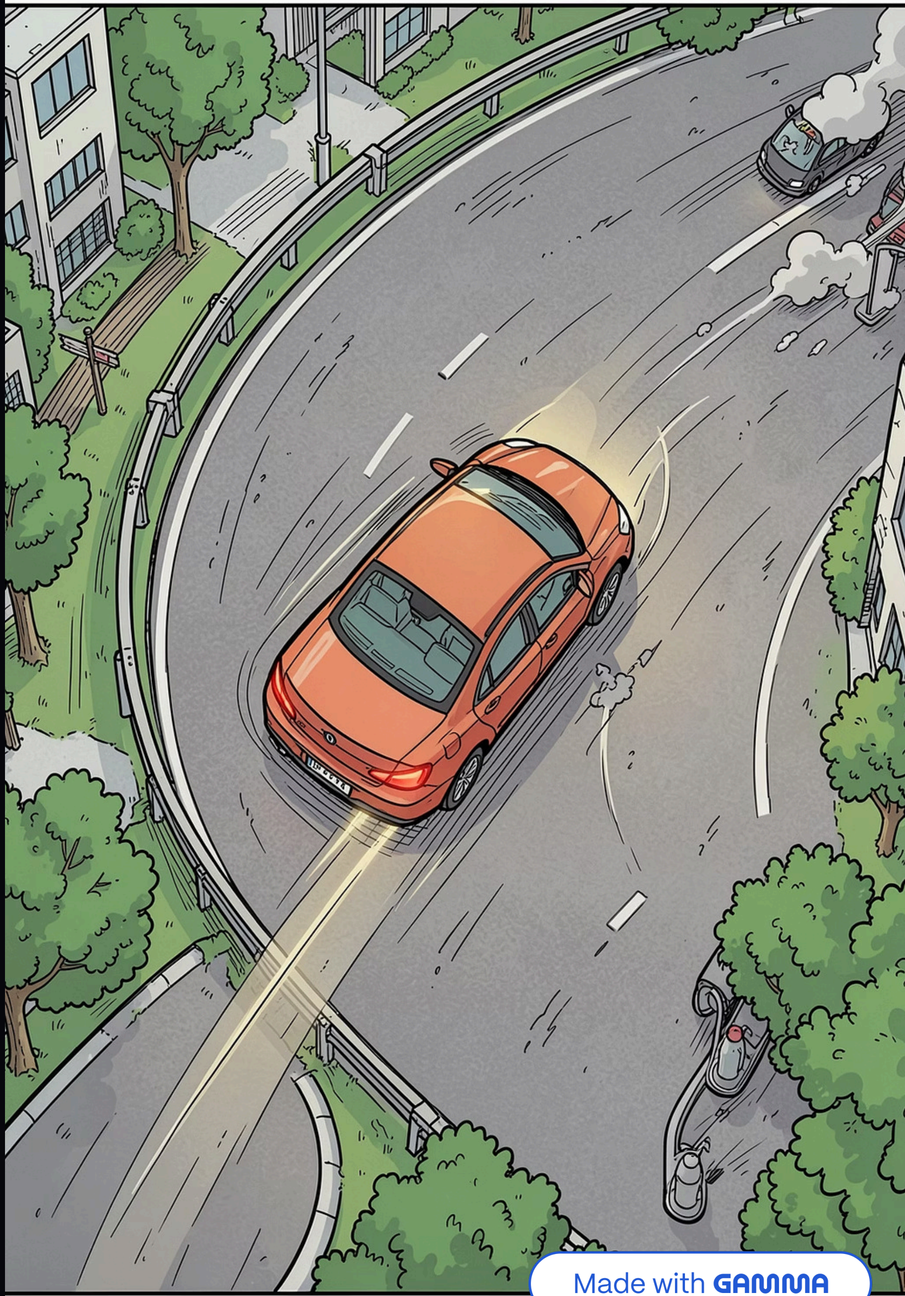
Roads are banked to provide centripetal force without relying solely on friction.

$$\tan\theta = \frac{v^2}{rg}$$

Maximum Safe Speed

On a flat road with friction:

$$v_{max} = \sqrt{\mu r g}$$



QUICK REFERENCE

Important Formulas at a Glance

Concept	Formula	SI Unit
Force (Newton's 2nd Law)	$F = ma$	Newton (N)
Momentum	$p = mv$	kg • m/s
Impulse	$J = F \times t = \Delta p$	N • s
Static Friction (max)	$f_s = \mu_s N$	Newton (N)
Kinetic Friction	$f_k = \mu_k N$	Newton (N)
Centripetal Force	$F_c = mv^2/r$	Newton (N)
Banking Angle	$\tan \theta = v^2/rg$	Degree / Radian

KEY TAKEAWAYS

Chapter Summary

1 **Newton's Three Laws**

Inertia → F=ma → Action-Reaction form the core of classical mechanics.

2 **Momentum is Conserved**

In isolated systems, total momentum before = total momentum after collision.

3 **Friction Opposes Motion**

Static > Kinetic > Rolling. Friction depends on normal reaction, not surface area.

4 **Circular Motion Needs Centripetal Force**

Banking of roads and friction together provide the required centripetal force.

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